

Behaviour identifies a low-dimensional parameter subspace tiled by the behavioural repertoire

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Abstract

Whole-organism biophysical simulators reproduce animal behaviour from hundreds to thousands of internal parameters, yet how many of those parameters behaviour actually constrains is unknown. Here we measure the behavioural-equivalence manifold, the set of parameter configurations that yield indistinguishable behaviour, across simulators spanning three phyla. We find that behaviour constrains a low-dimensional stiff subspace and leaves the remaining directions free, and we characterise that subspace in three ways. Its identity: in the nematode it is the neural substrate of the animal’s four-dimensional postural repertoire, and two independently built models generate behaviour that occupies the measured eigenworm space. Its dependence on behaviour: distinct behaviours constrain complementary rather than overlapping subspaces, so that the identifiable dimension is the union over the behavioural repertoire and grows as behaviours are added, which we demonstrate in nematode, fly-larva and rodent models. Its generality: the same organisation, with network connectivity and slow membrane properties stiff and fast voltage-gated inward currents sloppy, recurs in every system examined and in four external biological datasets that use none of our simulators. Behaviour therefore constrains whole-organism nervous systems on a low-dimensional subspace whose extent is set by behavioural richness, leaving an irreducible degenerate core.

Introduction

A central premise of mechanistic whole-organism simulators is that fitting them to behaviour recovers the biophysics that produces it. Decades of work on conductance-based models, however—from the stomatogastric ganglion (1, 2) to the broader theory of sloppy models (3, 4)—establish that many parameter sets produce indistinguishable dynamics: the map from parameters to behaviour is many-to-one (5–9). Whether, and how strongly, this degeneracy holds at the scale of a *whole-organism* nervous system—thousands of synaptic, junctional and channel parameters driving a behaving body—has not been measured. The consequences are direct: if behaviour constrains only a low-dimensional subspace, then behaviour-based calibration can recover that subspace but not the remainder, and the parameters it leaves unconstrained are precisely those that biology is free to vary.

Here we measure the behavioural-equivalence manifold of four published whole-organism motor simulators (BAAIWorm (10), modWorm (11), flybody (12), and NeuroMechFly v2 / FlyGym (13, 14)), which are built on physics engines for musculoskeletal control (15, 16). To test generality, we extend the same measurement across three phyla by adding a fly-larva model, a mammalian model, a connectome-constrained *Drosophila* visual circuit and the canonical crustacean pyloric population. We do not attempt to recover the true parameters; rather, we measure the manifold

of behaviourally equivalent ones, identify the constrained (stiff) subspace, and ask whether its low-dimensional geometry recurs across systems and in biology. The contribution is a measurement and its external validation.

The components required to do this at scale are now available. Connectome-constrained whole-organism simulators rest on increasingly complete wiring diagrams (17–24) on community simulation frameworks (25–28), and on the whole-brain neural recordings they aim to reproduce (29–31); high-throughput tracking reduces behaviour to low-dimensional kinematic descriptors (32–39); and parameter inference is increasingly amortised by simulation-based methods (40–44) or performed by black-box optimisers (45–48). What none of these components settles is *how much* of a whole-organism parameter set behaviour actually determines—the classical identifiability question (49, 50) that sloppy-model theory answers for small systems (3, 4, 51, 52) and that pyloric degeneracy answers for a single circuit (1, 2). Quantifying it at whole-organism scale is a prerequisite for behaviourally grounded digital organisms (53–56).

Results

Behaviour converges while parameters remain dispersed

We first asked whether the manifold has positive dimension. Repeating optimisation from independent starting points against a single fixed behavioural target, the behavioural loss converged while the recovered parameter vectors remained dispersed. On the 126-dimensional FlyGym controller, 29 of 30 runs reach the behavioural target while the solution cloud occupies an effective 21.7 of 126 dimensions; on the full coupled real modWorm model (OpenWorm connectome, 279 neurons), the behavioural curvature concentrates on 2 of 7 mechanism directions (Hessian analysis below), leaving the remainder unconstrained. Behavioural convergence with persistent parameter dispersion is direct evidence that the equivalence manifold has positive dimension (Fig. 1a).

The behavioural-equivalence manifold is low-dimensional

We next quantified the dimensionality directly from the behavioural-loss Hessian at the optimum. On the 767-parameter flybody policy, a Lanczos spectrum yielded 6 stiff directions at 90% and 11 at 99% of the spectral mass, with 28 of 40 sampled directions flat to $|\lambda| \sim 10^{-11}$. On the full coupled real modWorm model, the per-mechanism behavioural Hessian had effective dimension 2 of 7 (synaptic, gap-junction and synaptic rise-time directions stiff; capacitance and gain sloppy). On the 48-dimensional FlyGym walking controller, the behavioural Hessian had effective dimension 3 (90%) / 6 (99%) of 48 and spanned 19.5 orders of magnitude. On the BAAIWorm five-mechanism probe, the synaptic, gap-junction and motor-output scalings carried the curvature while the fast single-cell conductances did not. Across parameter counts of 5, 7, 48 and 767, behaviour is set by a low-dimensional stiff subspace embedded in a high-dimensional flat manifold (Fig. 1b).

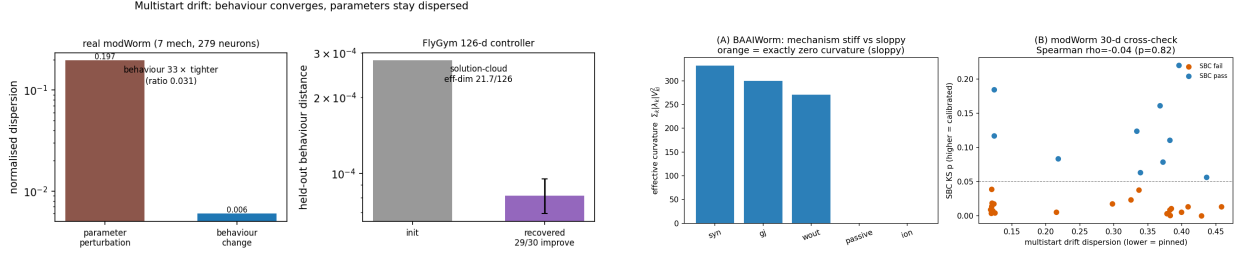


Figure 1: **The behavioural-equivalence manifold, measured.** **a**, Multistart drift: independent optimisations converge in behaviour while the recovered parameters remain dispersed (FlyGym 29/30 behavioural success, solution-cloud effective dimension 21.7/126; real modWorm behaviour approximately 30 $\times$ less variable than its parameters), indicating a positive-dimensional manifold. **b**, Behavioural-loss curvature at the optimum (BAAIWorm five-mechanism probe): the synaptic, gap-junction and motor-output axes carry the curvature (331.97/299.64/270.50), whereas the ion-channel and passive-membrane axes are behaviourally flat. A small number of stiff directions and many sloppy ones.

The manifold is the neural control of eigenworm posture

The nematode behavioural manifold is not an abstract low-dimensional set but the parameter-space image of the animal’s known approximately four-dimensional postural repertoire. Real *C. elegans* posture from the OpenWorm Movement Database occupied an effective 3.79–4.14 dimensions, with four eigenworms capturing 94–96% of postural variance, reproducing the classical eigenworm decomposition (32). Two independently constructed worm models generated behaviour occupying this same empirical eigenworm space: the modWorm posture subspace aligned with the real eigenworm basis at $\cos 0.77$ and BAAIWorm at $\cos 0.68$ – 0.69 (67–72% of variance). These alignments substantially exceeded those of random four-dimensional subspaces (modWorm $p=0.001$, BAAIWorm $p=0.04$ even within a conservative ten-dimensional posture ambient; $p<10^{-4}$ in the full space; Supplementary §6.1), and within each model it was the stiff mechanisms, not the sloppy ones, that displaced the eigenworm trajectory. Because the behaviour is approximately four-dimensional, it can constrain only approximately four parameter combinations; this is a property of the behaviour rather than of the sampling, as re-measuring with 7,200–14,400 posture observables left the effective dimension at 2–3 of 7 (Supplementary §6.2). The low-dimensional behavioural-equivalence manifold is thus the neural control of the animal’s low-dimensional posture (Fig. 2a).

Distinct behaviours constrain complementary subspaces

A low-dimensional behaviour raises the question of how a whole nervous system could be identified at all. The resolution is that distinct behaviours do not constrain the *same* few directions but complementary ones. In BAAIWorm, perturbing the chemosensory parameters altered behaviour 7.15 $\times$ more during chemotaxis (food-directed; sensitivity 0.872) than during plain locomotion in a weak gradient (0.122), whereas perturbing the motor parameters showed the opposite contrast (0.109 versus 0.357): the axis a behaviour renders stiff is the axis its function requires, and chemotaxis and locomotion render *different* axes stiff. This is consistent with the anatomical separation of the chemosensory navigation circuit from the ventral-cord locomotor circuit (57, 58), and with models in which distinct premotor and motor elements generate the locomotor rhythm (59–64). The identifiable subspace is therefore not the few directions of any single behaviour but the union over the behavioural repertoire. Within a single worm model this union increased as behaviours

were added: four distinct locomotor presets each constrained 2–4 of 7 mechanism directions with only partial overlap (mean pairwise stiff-subspace $\cos 0.54$); their behaviour-specific stiff directions spanned an effective 3.0 of 4 dimensions (a single shared direction would yield 1), each behaviour contributing on average a 0.54 component orthogonal to the others, so that adding a behaviour enlarges the identifiable subspace by a predictable increment (Supplementary §6.4). Their union reached 5; across twelve behaviours it saturated at 4 of 7, leaving a residual 2–3 directions that no behaviour constrains, an irreducible degenerate core. The stiff and sloppy directions are distributed parameter combinations rather than single parameters (stiff-subspace participation 3.42, sloppy 4.68 of 7), so this core is a genuine biological neutral space and not an unused parameter (Fig. 2b,c).

The same geometry recurs across three phyla

This low-dimensional, complementarily tiled organisation is not specific to the nematode. We measured it also in a fly larva (larvaworld) and a mammal (Virtual Rodent (65, 66), `extttm_control` (67, 68)), and in each case the identifiable dimension increased as distinct behaviours were added rather than being fixed by any one behaviour. In the larva, four genuinely distinct sensory behaviours (free exploration, chemotaxis, anemotaxis and feeding) each engage a different module—olfactory, wind-sensory and feeding—so that their union (8 of 10) was approximately double the single-behaviour dimension (3–5). In the rodent, a six-behaviour repertoire drove a still-increasing identifiability curve ($3.7 \rightarrow 12$ of 38 actuators). The absolute counts are not comparable across species, as the three models are parameterised differently and the rodent is driven open-loop, but the structure is invariant: a low-dimensional stiff subspace per behaviour, complementary across behaviours, with an identifiable dimension set by behavioural richness. The crustacean pyloric population and a *Drosophila* visual circuit (below) extend the same stiff/sloppy geometry to a fourth and fifth system (Fig. 2d).

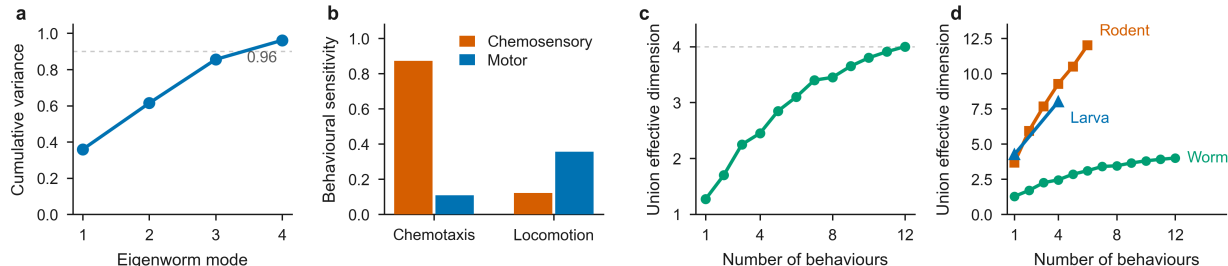


Figure 2: The identifiable subspace is the eigenworm space, tiled complementarily by behaviour, across phyla. **a**, Cumulative postural variance of real *C. elegans* posture (OpenWorm Movement Database, $n=8432$ frames) across eigenworm modes; four modes capture 0.96 of the variance (dashed line, 0.90). Two independently constructed worm models align with this empirical basis at $\cos 0.77$ (modWorm) and $\cos 0.68$ (BAAIWorm). **b**, Behavioural sensitivity to perturbation of the chemosensory (vermillion) and motor (blue) parameter classes, during food-directed chemotaxis and during plain locomotion in a weak gradient. Chemosensory parameters are stiff during chemotaxis but sloppy during locomotion (a $7.15\times$ context shift) and motor parameters show the reverse contrast, so distinct behaviours constrain distinct axes. **c**, Union effective dimension of the identifiable subspace as behaviours are added (real modWorm, twelve behaviours), saturating at 4 of 7 mechanism directions (dashed line) and leaving a residual degenerate core. **d**, The same increase with repertoire size in three phyla: nematode (green, 7 mechanisms), fly larva (blue, 10 parameters) and rodent (vermillion, 38 actuators). Absolute counts are not comparable across species because the models are parameterised differently; the structure is invariant.

Two independent probes agree on the identifiable axes

Computed on the full coupled real modWorm model within a single shared prior region (per-mechanism log-multipliers, prior scale 0.15), the two geometric probes concurred on which axes are identifiable: per-mechanism Hessian stiffness and negative manifold drift were rank-correlated at Spearman $\rho=0.96$ ($p=5\times 10^{-4}$, 7 mechanisms). A third probe served as a calibration control rather than a parallel identifiability ranking: an amortised neural posterior (NPE/MAF, `sbi`) trained in the same region on $N=300$ real simulations was well-calibrated on all 7 mechanisms (per-axis simulation-based calibration passes 7/7; Kolmogorov–Smirnov $p > 0.05$). This confirms that the inference is sound but, because a calibrated posterior calibrates every axis, does not by itself yield a per-axis stiff/sloppy ordering. We therefore read the manifold’s stiff/sloppy geometry from the two probes that agree, using simulation-based calibration as a control on the inference.

Stiff and sloppy directions are biophysically interpretable

At the BAAIWorm optimum the stiff axes were the synaptic (331.97), gap-junction (299.64) and motor-output (270.50) weights. A finer per-channel decomposition over 20 named mechanisms (Supplementary §5.2) resolves the partition further: the stiffest axes are the passive leak, the gap-junction and synaptic weights, the motor output and several potassium conductances, whereas the fast voltage-gated calcium channels EGL-19 and UNC-2 and the calcium-activated potassium channels SLO-1/SLO-2 are the sloppiest. Behaviour is thus determined by how cells are connected and by their passive and slow properties, and is largely insensitive to the fast voltage-gated calcium conductances, consistent with the known compensation and variability of these currents in small circuits (2). A complementary probe over the full 3076-weight connectome found the connection weights collectively stiff (Supplementary §5.3), locating the degeneracy in the single-cell biophysics

rather than in the connectivity. This partition is not an artefact of one model: probed identically, two independently constructed worm simulators—modWorm (coupled-ODE, 7 mechanisms) and BAAIWorm (NEURON, 20 named channels)—agree that the network-connectivity axes are stiff (stiff fraction 0.83 / 0.87) while the single-cell biophysical axes are sloppy (0.08 / 0.16), indicating that “connectivity stiff, single-cell sloppy” is a property of the organism rather than of a particular implementation.

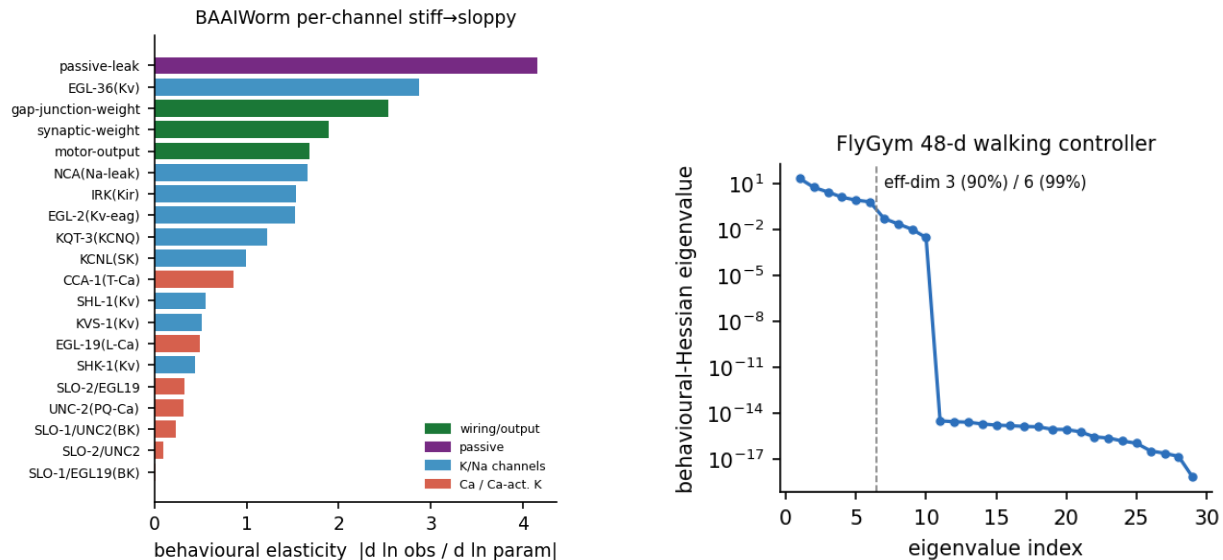


Figure 3: **Stiff/sloppy structure at named-channel and higher-dimensional scale.** Left: BAAIWorm per-channel behavioural elasticity over 20 named mechanisms, ordered stiff (top) to sloppy (bottom); the connectivity (gap-junction, synaptic and motor) and passive-leak axes are stiff, and the fast voltage-gated calcium channels EGL-19/UNC-2 and the calcium-activated potassium channels SLO-1/SLO-2 are sloppy. Right: behavioural-Hessian eigenspectrum of the 48-dimensional FlyGym walking controller, effective dimension 3–6 of 48.

The same geometry appears in external biological data

Four external datasets, none of which involves our forward models, exhibited the same low-dimensional stiff/sloppy structure (Fig. 4). (1) *Pyloric degeneracy*. Applying our Hessian probe to the Prinz–Marder population of conductance-based models that produce the same triphasic pyloric rhythm (1) (rhythm-validity gated; valid rate 0.56%) yielded a 31-parameter eigenspectrum spanning approximately ten orders of magnitude with effective dimension 2.13/31—only 2–6 stiff directions, with stiffness on the leak and inhibitory-synapse conductances and sloppiness on the fast sodium conductances, the currents known to be compensated in this circuit. This canonical degeneracy dataset exhibits the same low-dimensional stiff/sloppy partition as the simulators. (2) *Neuronal property variability*. Across $n=1914$ Allen cortical neurons, electrophysiological properties span a stiff/sloppy gradient (coefficient of variation 0.08–1.85: resting potential stiff, adaptation and firing rate sloppy) on a property-space manifold of participation dimension 4.78/11. (3) *Gene conservation*. Across 40 *C. elegans* ion-channel and synaptic genes, ortholog identity across *Caenorhabditis* ranges from 77.9 to 99.9%; the gap-junction gene *unc-9* is the single most conserved (99.9%), paralleling the stiffness of the gap-junction axis in (1). (4) *Behavioural variation*. Across $n=61$ isogenic N2 worms (683 Tierpsy features), cross-individual behaviour occupies a participation dimension of

9.75; genetically identical animals vary on an approximately ten-dimensional behavioural manifold, the empirical counterpart of the parameter manifold. The directions that biology conserves (stiff) and permits to vary (sloppy) coincide with the simulators' identifiable and degenerate axes.

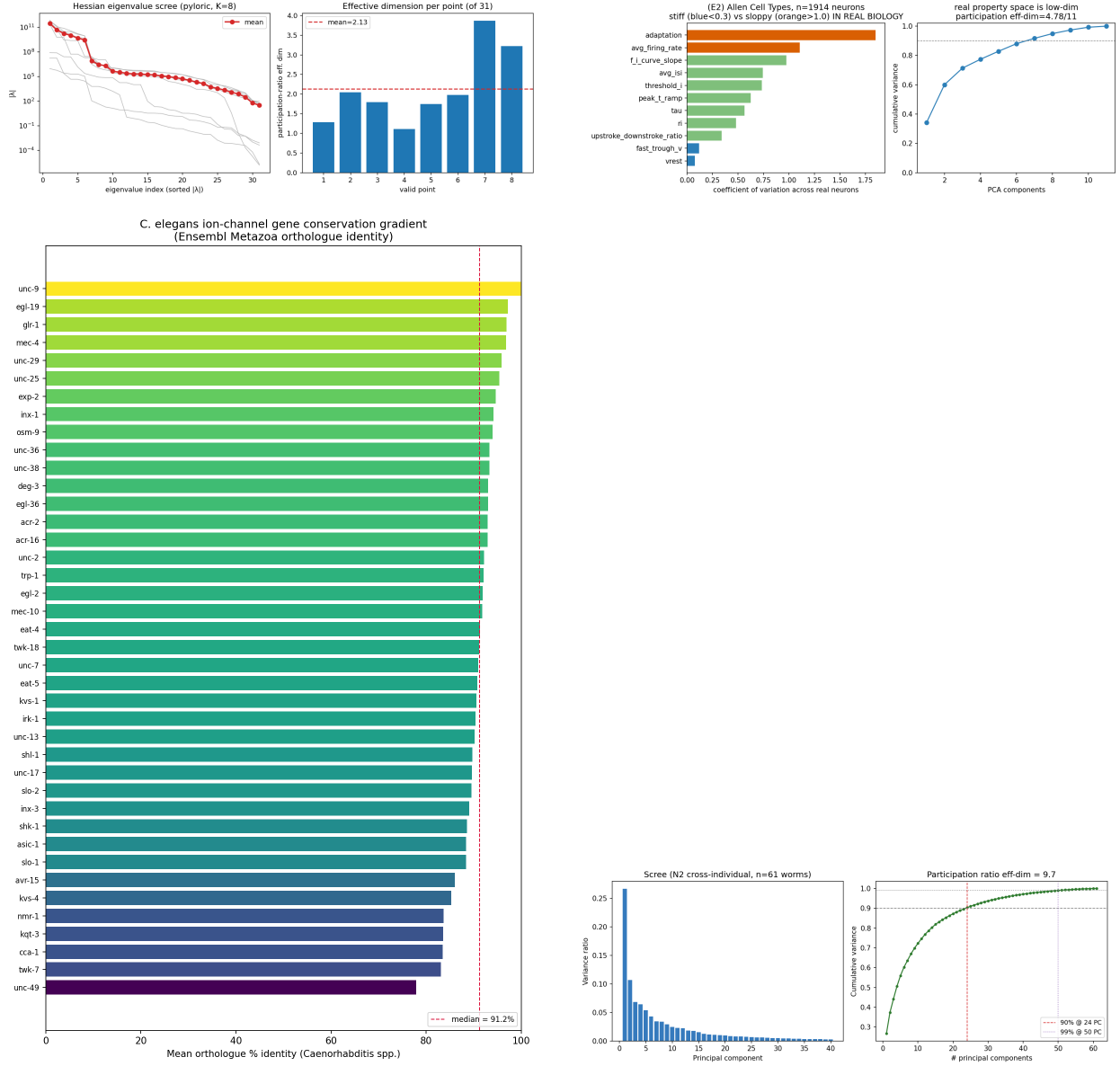


Figure 4: The same low-dimensional stiff/sloppy geometry in four biological datasets that do not use our simulators. Top-left: Prinz–Marder pyloric model population, rhythm-gated, 31-parameter Hessian eigenspectrum (effective dimension 2.13/31; stiff = leak and inhibitory synapses, sloppy = fast sodium). Top-right: Allen cortical-neuron property variability ($n=1914$; stiff/sloppy coefficient-of-variation gradient, participation dimension 4.78/11). Bottom-left: *C. elegans* ion-channel-gene conservation across *Caenorhabditis* ($n=40$ genes; *unc-9* gap junction most conserved). Bottom-right: isogenic-N2 behavioural manifold (OWMD, $n=61$ worms, 683 Tierpsy features; participation dimension 9.75).

Discussion

Whole-organism nervous systems are extensively non-identifiable from behaviour, but the non-identifiability is structured, and the structure has a simple organising principle: what behaviour can identify is set by how rich the behaviour is. Any single behaviour is low-dimensional—in the nematode, the approximately four-dimensional eigenworm space—and constrains only a few stiff parameter combinations; but distinct behaviours constrain complementary combinations, those that their particular function requires (chemotaxis the chemosensory axes, locomotion the motor axes), so that the identifiable subspace is the union over the behavioural repertoire and increases with the repertoire. This recasts degeneracy as a measurable resource rather than a fixed limit: a connectome-constrained simulator is identifiable up to the diversity of behaviours one is willing to elicit and measure, and the directions that remain after every behaviour has been used—the residual degenerate core—are the parameters that biology is free to vary. The result lifts the sloppy-models picture (3, 4, 51) and pyloric degeneracy (1, 7) from single circuits to behaving whole organisms, and it explains why behaviour-based calibration recovers a working parameter set without recovering unique biophysical values: it recovers the stiff union, not the core. The unconstrained directions are not noise; the external datasets show that they are also the more variable and less conserved directions in biology (cortical-property variability; cross-*Caenorhabditis* gene conservation), consistent with the biophysical diversity documented within identified cell types (69, 70), so that the manifold is a map of which mechanisms behaviour can and cannot identify. The same principle accounts for the one apparent exception: applying the probe to a connectome-constrained *Drosophila* visual model (Supplementary §5.7) yields a high-dimensional manifold (effective dimension 43 of 65 cell types), because a precision sensory code constitutes a far richer behaviour than locomotion; yet the invariant holds, its stiff axes being precisely the load-bearing motion-detector pathway (T4/T5). The dimension of the manifold scales with the behaviour against which it is measured, and which mechanisms are stiff is always determined by what the behaviour requires.

Limitations. The per-channel decomposition (Supplementary §5.2) reaches named single conductances but still scales each conductance type uniformly across cells rather than cell by cell; the mapping from parameter axes to biophysical classes is biophysically informed rather than learned; and the alignment between simulator stiff axes and the external biological datasets (1)–(4) is established at the level of low-dimensionality and qualitative direction rather than per-eigenvector correspondence. The behaviour-to-neural prediction that would close the loop is confirmed only qualitatively—behaviour-recovered parameters reproduce the gross statistics (dimensionality, correlation scale) of the unobserved neural state, but in the modality of membrane voltage compared against immobilised calcium imaging (Supplementary §5.4), not as a per-neuron match. An external check of the stiff/sloppy ordering against real mutant phenotypes remains data-limited (Supplementary §5.6).

Methods

Manifold drift. For each simulator, N optimisations (separable CMA-ES or SPSA) from independent starts are fitted to one fixed behavioural target; we report behavioural recovery, the normalised per-dimension dispersion of the N recovered parameter vectors, and the PCA participation dimension of the solution cloud. **Hessian.** The behavioural-loss Hessian at the optimum is computed by finite differences (low dimension) or Lanczos iteration (high dimension); the effective dimension is the number of directions accounting for 90/99% of the spectral mass, and a direction is flat when

$|\lambda|$ reaches the finite-difference floor. Its Gauss–Newton form is the Fisher information matrix of the parameters given the behavioural observables, so its effective dimension is the local practical-identifiability rank in the standard sense (3, 49). We adopt it as the primary cross-simulator metric because it is robust to the choice of observable and parameterisation, unlike a raw perturbation-sensitivity ratio, and because the manifold-drift probe provides an independent, optimisation-based cross-check that agrees with it ($\rho=0.96$). **Eigenworm identity.** Real *C. elegans* skeletons (OWMD N2) and model body postures are reduced to tangent-angle eigenmodes by PCA; we report the posture effective dimension, the top-four cumulative variance, and the principal-angle cosine between model and real eigenworm subspaces, with significance assessed against random subspaces of matched dimension. **Complementary tiling and saturation.** Per-behaviour stiff subspaces are taken as the leading Hessian eigenvectors under each behavioural target; we report pairwise stiff-subspace cosines, the union effective dimension over a behaviour set, and the saturation curve of union dimension against the number of behaviours. Context dependence (BAAIWorm) is the ratio of a parameter class’s behavioural sensitivity between two behaviours (chemotaxis versus weak-gradient locomotion). The same union/saturation construction is applied across species (fly larva, larvaworld; rodent, Virtual Rodent / dm_control). **Calibration.** Per-dimension rank-based simulation-based calibration (71, 72) of an amortised neural posterior (73, 74); a dimension passes when the per-axis Kolmogorov–Smirnov test is non-significant at $\alpha=0.05$. **External datasets.** The Prinz–Marder pyloric model population re-gated to the triphasic rhythm and probed with the identical Hessian-eigenspectrum method; Allen Cell Types electrophysiology; WormBase ParaSite / Ensembl Metazoa ortholog identities; and OpenWorm Movement Database isogenic-N2 Tierpsy features. All external data are public, and no simulator is used in datasets (1)–(4). Full specifications, cohort sizes and reproduction commands are given in the Supplementary Information.

Data availability

All datasets analysed in this study are publicly available. The external biological references are: the Prinz–Marder conductance-based model population of the stomatogastric ganglion (1); the Allen Cell Types electrophysiology database (<https://celltypes.brain-map.org>); ortholog identities from WormBase ParaSite and Ensembl Metazoa (<https://parasite.wormbase.org>); and the OpenWorm Movement Database (<http://movement.openworm.org>). The whole-organism simulators are the published releases of BAAIWorm (10), modWorm (11), flybody (12), NeuroMech-Fly v2 / FlyGym (13), the connectome-constrained optic-lobe model of Lappalainen et al. (75), larvaworld, and the Virtual Rodent of the dm_control suite. All quantities reported in the main text and Supplementary Information are provided as result files in the accompanying deposit, one file per analysis as named in each Supplementary section.

Code availability

All analysis code, the per-experiment result files, and a container that reproduces every reported quantity and figure from those files are provided in the accompanying deposit. The reproduction entry point verifies each numerical claim against its source file and regenerates the figures.

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Author contributions

C.H. conceived the study, designed and performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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